

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Led **electrical design and power architecture validation** across feasibility, concept, scheme, and detailed design phases for mission-critical infrastructure projects, defining utility intake, **MV/LV distribution, UPS topologies**, generators, switchgear, PDUs, STS, busway, earthing, and lighting strategies aligned to **Tier II–IV availability objectives**; coordinated multi-disciplinary interfaces with mechanical, structural, architectural, fire/life safety, and ICT teams to deliver fully integrated designs supporting 99.99% uptime SLAs.
- Performed **power systems studies and protection design** using **ETAP/SKM tools** including load flow analysis, short-circuit studies, protection coordination, arc-flash hazard analysis, and harmonic analysis; assured compliance with **IEC/BS/EN and NEC/NFPA standards**, client specifications, and AECOM quality procedures across UK and EMEA project portfolios.
- Integrated **monitoring and controls systems (BMS/EPMS/SCADA)**, metering strategies, and resilience features; designed and validated **advanced control systems** and **grid-interactive strategies** supporting real-time operational visibility, reducing incident response time by **40%** and improving system availability by **3.2%** across enterprise infrastructure.
- Embedded **decarbonization and energy efficiency** measures including **high-efficiency UPS systems**, advanced controls optimization, heat recovery integration, demand response capabilities, and on-site renewable energy design; delivered **15% operational energy reduction** through innovative architectural and electrical efficiency strategies.
- Led **hyperscale and multi-site program delivery** across fast-track and design-build environments; managed project work streams across UK and EMEA geographies, coordinating procurement support, construction oversight, commissioning protocols, and handover procedures; mentored teams of technical experts and demonstrated strong commercial acumen in managing project financials and stakeholder relationships.
- Demonstrated client-facing and work-winning capability through business development engagement with tech firms and mission-critical facility operators; leveraged detailed knowledge of **mechanical design engineering practices and data centre operational interfaces** to position technical excellence, quality assurance, and resilience outcomes; served as internal and external ambassador across EMEA data centre community.

SKILLS

Electrical Design & Power Systems – utility intake design, MV/LV distribution, UPS topologies, generator sizing, switchgear specification, PDU architecture, STS systems, busway design, earthing design, lighting design, Tier II–IV availability architecture.

Power Systems Analysis & Tools – ETAP, SKM, load flow analysis, short-circuit studies, protection coordination, arc-flash analysis, harmonic analysis, protection device selectivity.

Standards & Compliance – IEC 60364, IEC 61936, BS 7909, EN 50160, NEC (National Electrical Code), NFPA 110, NFPA 75, UL standards, client specifications, design code validation.

Controls & Monitoring Systems – BMS (Building Management Systems), EPMS (Energy and Power Management Systems), SCADA, metering strategies, power monitoring systems, grid-interactive controls, demand response integration.

Energy Efficiency & Decarbonization – high-efficiency UPS design, advanced control optimization, heat recovery systems, demand response strategies, on-site renewable energy integration, energy modelling, carbon

footprint reduction.

Project Delivery & Leadership – hyperscale program management, multi-site delivery, fast-track project execution, design-build environments, team mentoring, procurement coordination, construction oversight, commissioning, handover management, commercial project management.

Cross-Disciplinary Coordination – mechanical-electrical integration, structural coordination, architectural alignment, fire and life safety interfaces, ICT integration, security system coordination, business development, stakeholder management.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
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BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021
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PROJECTS

Hyperscale Data Centre Power Architecture & Resilience Programme (HCL Technologies)

- Owned end-to-end **electrical design and power architecture** for a **50+ MW hyperscale data centre deployment** across three geographically distributed UK sites; defined and validated utility intake strategies, dual-corded **MV/LV distribution networks**, redundant **UPS topologies (2N architecture)**, and distributed generator placement to achieve **Tier IV availability (99.995%)** and support 24/7 operations.
- Executed **power systems studies** using **ETAP 16.0** across all three sites including load flow analysis for 15,000+ circuit loads, three-phase short-circuit analysis, **protection coordination** with selectivity validation, **arc-flash hazard analysis** for PPE specification, and harmonic content assessment; ensured compliance with **IEC 61936, BS 7909, EN 50160, and NFPA 110** standards, reducing design rework by **28%**.
- Designed and integrated **SCADA/EPMS monitoring and BMS control systems** providing real-time visibility into 180+ electrical assets (switchboards, PDUs, UPS units, generators, STS systems); implemented **advanced metering strategies** and predictive alerting reducing unplanned downtime by **32 hours annually** and enabling **dynamic load balancing** across redundant power paths.
- Embedded **decarbonization initiatives** including **high-efficiency UPS systems (≥96% efficiency)**, waste heat recovery from generators for facility heating, **on-site 5 MW solar PV** integration with **grid-interactive inverters**, and demand response scheduling reducing grid energy draw by **18%** and carbon footprint by **2,340 tonnes CO₂e annually**.
- Coordinated multi-disciplinary design integration across mechanical (CRAC/CRAH placement, cooling ducting interfaces), structural (electrical room layouts, cable tray routing loads), architectural (customer suites, visitor pathways), fire/life safety (cable segregation, detection zones), and ICT (fibre routing, equipment racks); delivered **16 detailed design packages** with **zero compliance defects** at design review.

Mission-Critical Power Resilience & Client Business Development (HCL Technologies)

- Demonstrated **client-facing and work-winning capability** by leading technical presentations and feasibility studies with three tier-1 hyperscale cloud operators and two financial services firms; articulated **electrical resilience architectures**, decarbonization value propositions, and operational cost savings (average **£840k NPV over 10 years**), resulting in **£12M in follow-on contract wins**.
- Mentored a team of six electrical engineers and technicians across UK and EMEA sites; established design standards documentation, peer-review protocols, and technical capability frameworks; improved junior engineer **design autonomy by 65%** within 12 months through structured mentorship and training on **ETAP modelling, IEC 61936 standards, and Tier architecture design**.

- Managed project financials and commercial delivery across **£3.2M electrical scope**; tracked design labour, equipment procurement, and construction costs against budget with **98.4% cost accuracy**; negotiated supplier agreements for **UPS, switchgear, and generator packages** achieving **12% cost reduction** while maintaining **Tier IV specification compliance**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)